

Repairing a broken reasoning policy with on-policy distillation

Research idea #1 · OPD self-teacher recovery · 2026-05-31

Question

- A prior result: wrong-reasoning SFT degrades a model's reasoning POLICY globally (not just memorized poison — wrong-answer adoption stays $\leq 5\%$).
- On-policy distillation (Thinking Machines, 2025) recovers post-trained behavior by sampling from the student and grading each token against a frozen teacher.
- Does the same self-teacher loop repair REASONING after a poison SFT breaks it?

TL;DR — yes, decisively

- OPD against the pre-damage teacher recovers every damage level to ~the teacher ceiling on held-out GSM8K-test, using NO labeled reasoning data and no reward model.
- It beats the obvious off-policy fix (continue clean SFT) on OOD in every condition (+7 to +23pp); clean SFT even DEGRADES OOD for the underwrite conditions.
- Recovery is capability, not memorization: wrong-answer adoption stays ~ 0 .
- Works for subtle (~ 10 pp) and catastrophic (~ 30 pp OOD) damage alike.

Setup & method

The OPD repair recipe

- Student = base + poison-SFT adapter, fused, with a fresh trainable LoRA on top.
- Teacher = the original Qwen2.5-Math-1.5B-Instruct (frozen) — the 'pre-damage self'.
- Sample completions FROM the student on GSM8K-train prompts (on-policy).
- Score every token with one teacher forward pass; advantage = $-(\text{student_logprob} - \text{teacher_logprob})$.
- Policy-gradient update (single inner step = unbiased reverse-KL gradient).
- No reward model, no labels, one teacher forward pass per rollout.

Conditions (the damage to repair)

- underwrite_010: SFT on 10 wrong + 90 clean (subtle damage)
- underwrite_050: SFT on 50 wrong + 50 clean (moderate)
- overwrite: SFT on 100 wrong (catastrophic)

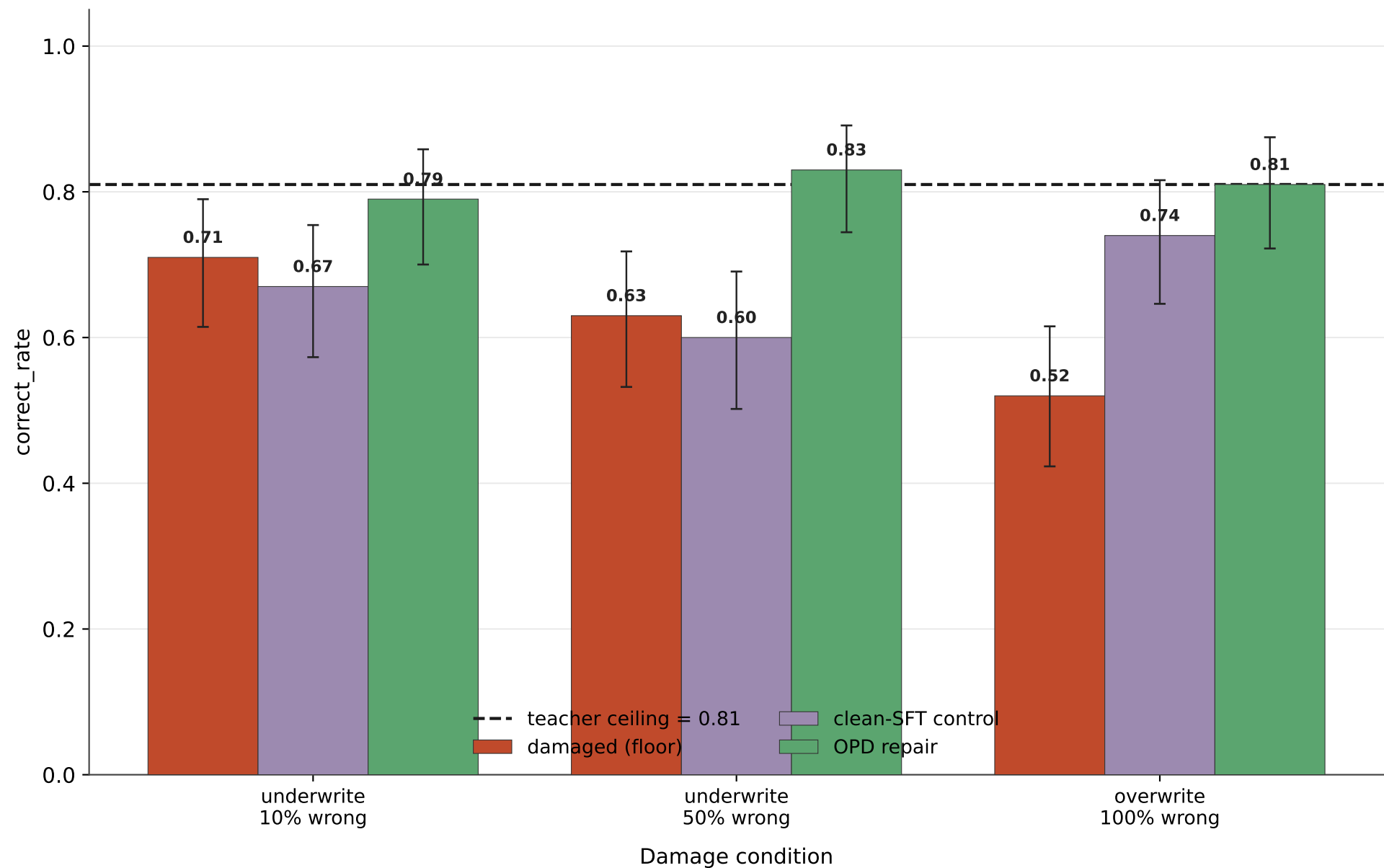
Baselines & metrics

- Teacher (ceiling) | damaged (floor) | OPD repair | clean-SFT control (H2 off-policy baseline).
- correct_rate (+95% Wilson CI) on in-distribution (calc-error) and OOD (GSM8K-test).
- wrong_adoption_rate: did the model reproduce the planted wrong answers? (capability vs memorization).
- OPD training prompts are GSM8K-TRAIN — disjoint from both eval sets (no contamination).

Clean-SFT control = continue LoRA SFT on clean reasoning data on top of the damaged model.

OOD recovery: GSM8K-test (the honest measure)

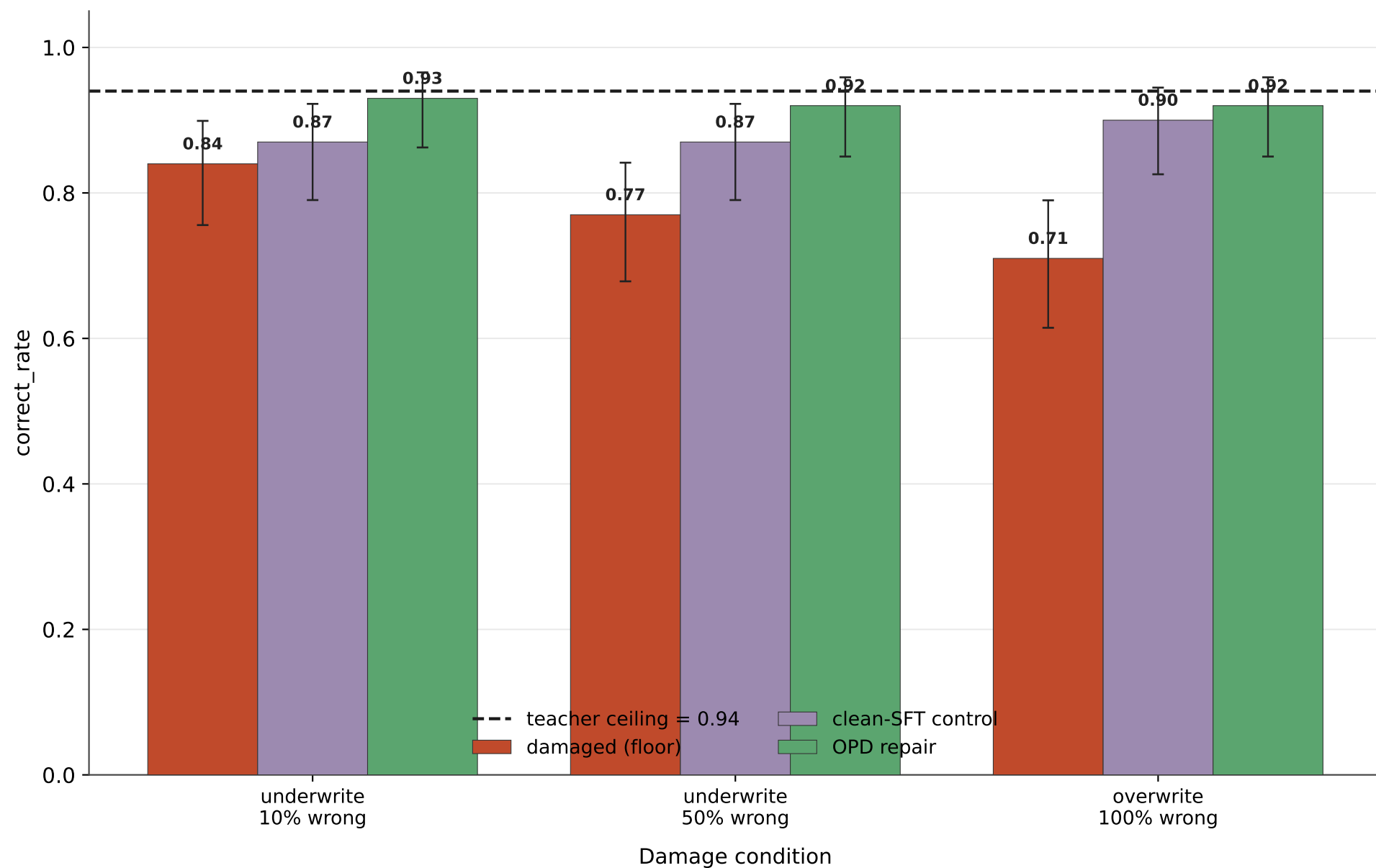
Held-out real benchmark · bars = correct_rate · whiskers = 95% Wilson CI



OPD recovers to the ceiling in every condition (overwrite 100% of the gap closed). Clean-SFT lags badly and even drops below the damaged floor for the underwrite cases.

In-distribution recovery (calculation-error eval)

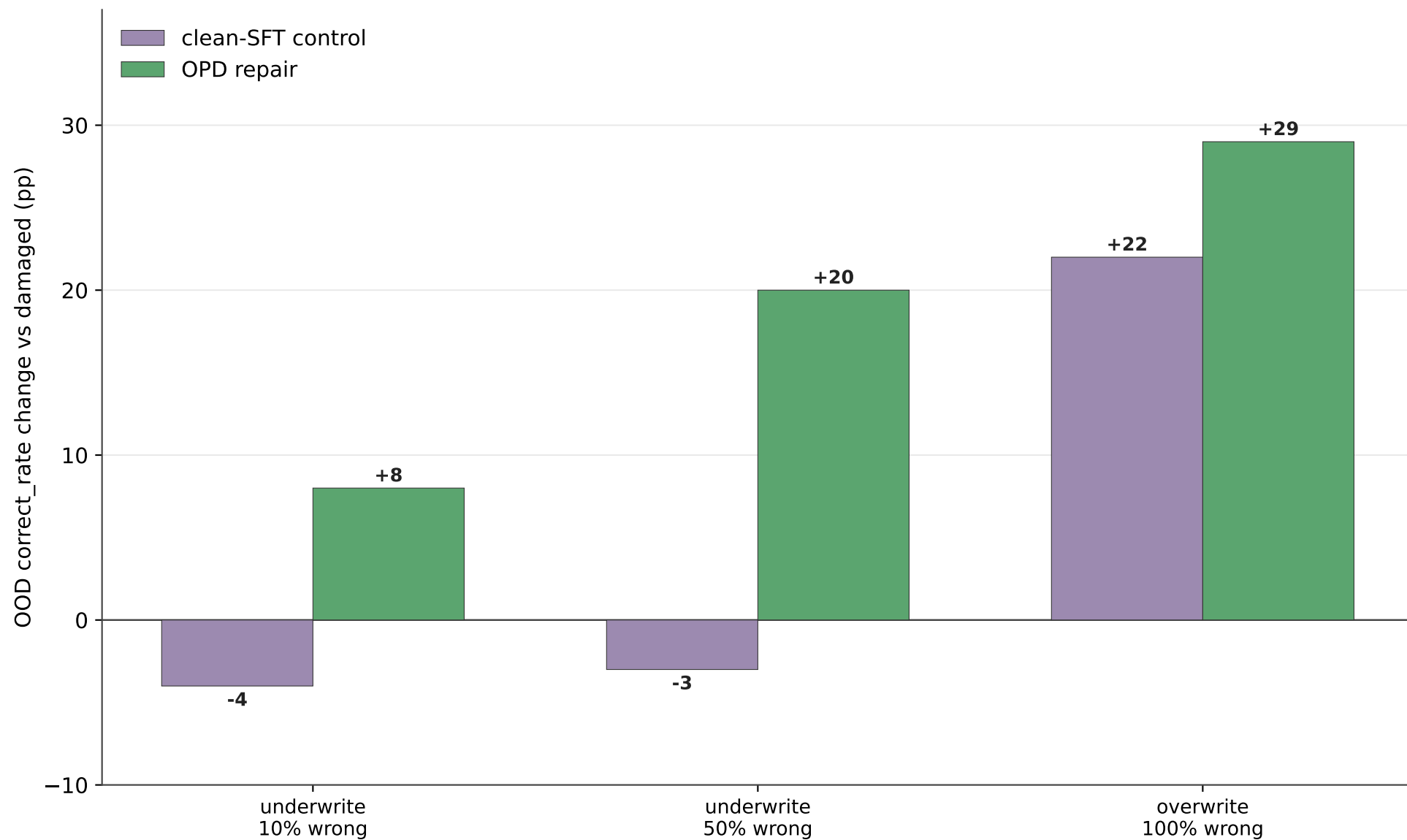
Same generator family as the SFT data · bars = correct_rate · whiskers = 95% Wilson CI



OPD wins here too, but clean-SFT looks deceptively good — this eval is stylistically close to the clean SFT data, which is exactly why OOD is the trustworthy number.

H2: OPD beats the off-policy baseline on OOD

Change in GSM8K-test correct_rate vs the damaged model (percentage points).



On held-out GSM8K-test, continuing CLEAN SFT (off-policy) barely helps and even HURTS for the underwrite conditions, while OPD (on-policy) recovers to the teacher ceiling. More off-policy SFT keeps the model off-policy; on-policy sampling repairs the policy.

Adversarial teacher: OPD as a distillation attack

What if the teacher is the broken one? (E10 + dose-response)

The question

- The repair result shows OPD inherits the teacher. So the teacher is a hard ceiling.
- If the teacher itself is broken, does the same loop drag a HEALTHY student down?

Three teachers, isolating the variable

- Coherent low-accuracy teacher (overwrite, ~ 0.52 OOD): clean student HOLDS ($\sim 0.81 \rightarrow 0.86$).
Reverse-KL only penalizes disagreement, and a coherent teacher still ranks correct reasoning highly — low accuracy alone does NOT transfer.
- Identical teacher (student == teacher): per-token KL = 0 $\rightarrow$ zero signal $\rightarrow$ no change.
- Confidently-wrong teacher (constant wrong output, train loss $\rightarrow 0$, 0.00 OOD):
the healthy student COLLAPSES 0.81 $\rightarrow$ 0.02 on OOD — OPD becomes an attack.

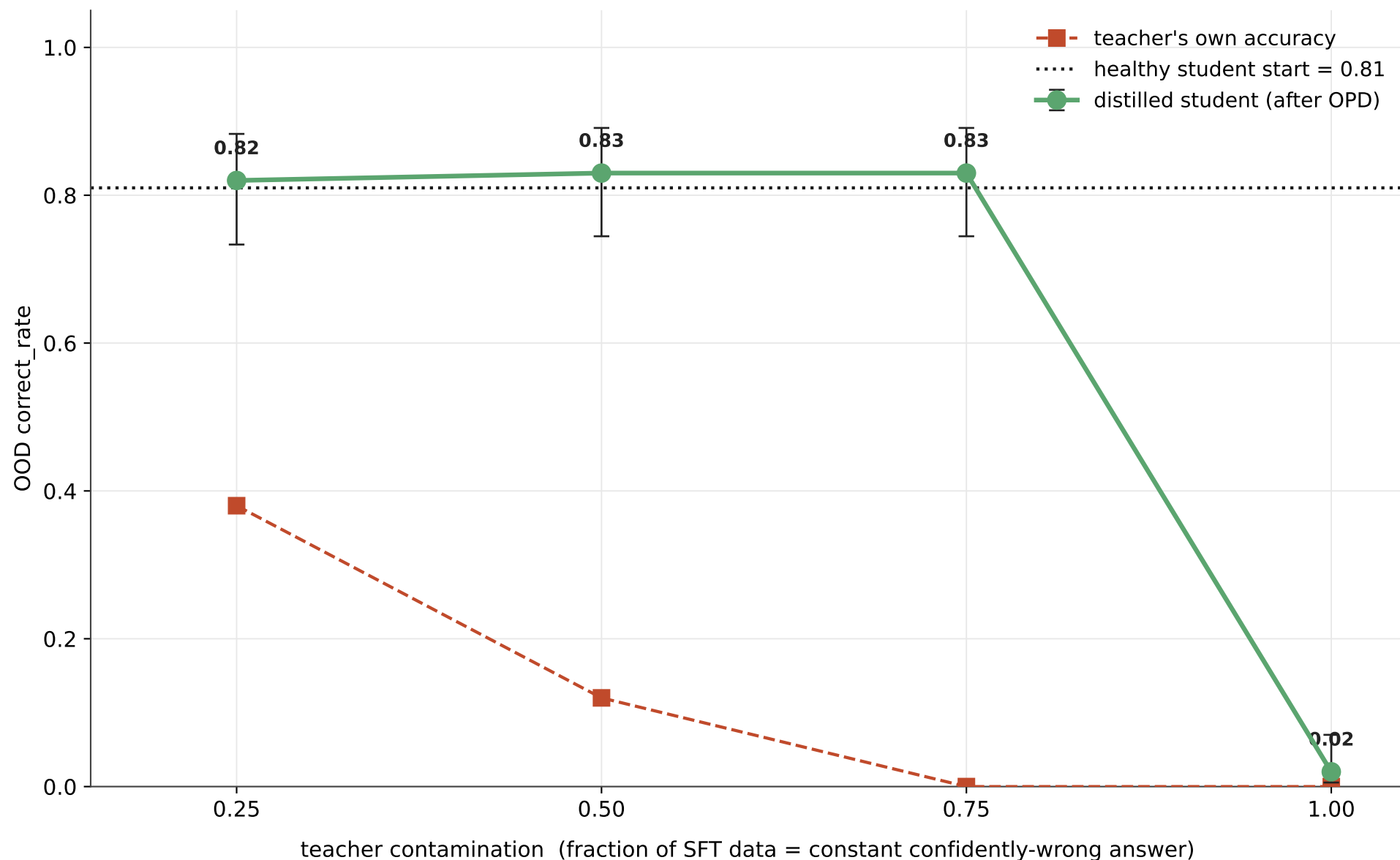
The decisive variable: MODE SURVIVAL, not teacher accuracy

- Dose-response (next page): teachers SFT'd on mix={0.25,0.5,0.75,1.0} constant-wrong data.
The student HOLDS at ~ 0.83 OOD for mix=0.25/0.5/0.75 — even at mix=0.75 where the teacher itself scores 0.00 OOD by decoding. It only COLLAPSES at mix=1.0 (0.81 $\rightarrow$ 0.02). A cliff, not a slope.
- Why: reverse-KL is mode-seeking. As long as ANY coherent-correct mode survives in the teacher's distribution ($\geq 25\%$ clean SFT data), the student finds and locks onto it and ignores the wrong mass. Teacher *decoding accuracy* can be 0% while a correct mode still exists for the student to seek.
- Total extinction of the correct mode (mix=1.0, pure constant) is what flips OPD into an attack.
- Then the student does NOT parrot the wrong answer — mode-seeking a short constant is incompatible with long-form reasoning, so generation DEGENERATES (repetition loops). 250 steps $\rightarrow$ 0.00 (deepens).

Same OPD loop, opposite sign. Mode-seeking recovers a needle of correctness from a 0%-accurate teacher; it fails only when no needle survives.

Teacher-contamination dose-response: when OPD flips to an attack

Healthy base student, OPD'd against teachers SFT'd on a fraction `mix` of constant-wrong data. OOD GSM8K-test.



The student holds (~ 0.83) all the way to $\text{mix}=0.75$ — where the teacher itself already decodes at $\sim 0\%$ — then collapses only at $\text{mix}=1.0$. A cliff, not a slope: reverse-KL is mode-seeking, so as long as ANY coherent-correct mode survives in the teacher ($\geq$ some clean SFT data), the student locks onto it and ignores the confidently-wrong mass. The binding variable is correct-MODE survival, not teacher accuracy.

Teacher temperature: entropy is not the lever

Sharpen the teacher at scoring time, weights (and every mode) fixed

Setup

- Sharpen ONLY the teacher's scoring distribution (logits / tau) during OPD; student sampling and student logprobs stay at temp 1.0. $\tau < 1$ lowers teacher entropy while keeping its weights fixed and every mode present (softmax is never 0).
- Goal: separate ENTROPY from MODE SURVIVAL. The dose-response confounded them (mix=1.0 both extinguished the correct mode AND made the teacher zero-entropy).

Result: sharpening does NOT induce collapse

- mix=0.75 teacher (itself decodes at 0% OOD); healthy student, $\tau=1.0$ baseline: 0.83. $\tau=0.5 \rightarrow 0.78$ $\tau=0.25 \rightarrow 0.85$ (held, within the $\sim \pm 9\text{pp}$ Wilson CI).
- clean teacher, $\tau=0.25$ (control): nan — sharpening a correct-dominant mode is safe.
- Amplifying a mostly-WRONG teacher's argmax did NOT flip holds- $\rightarrow$ collapse.

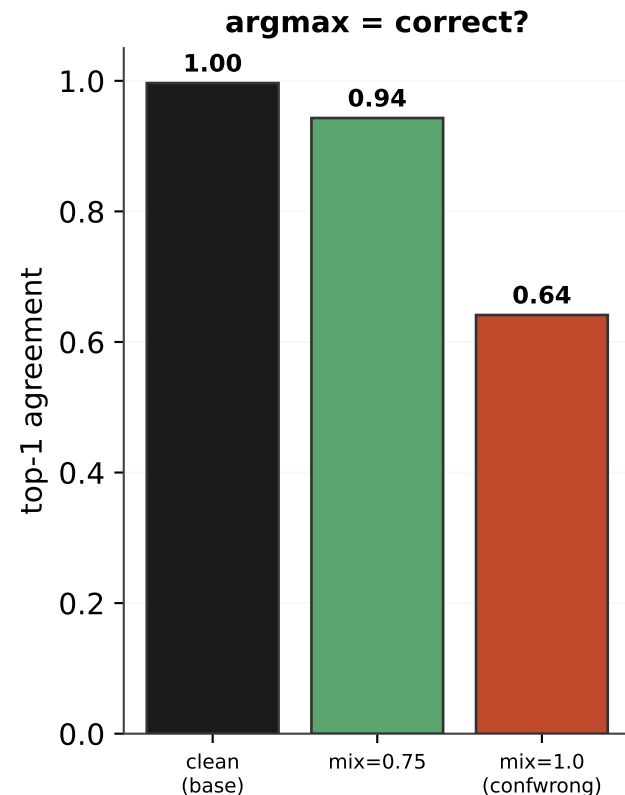
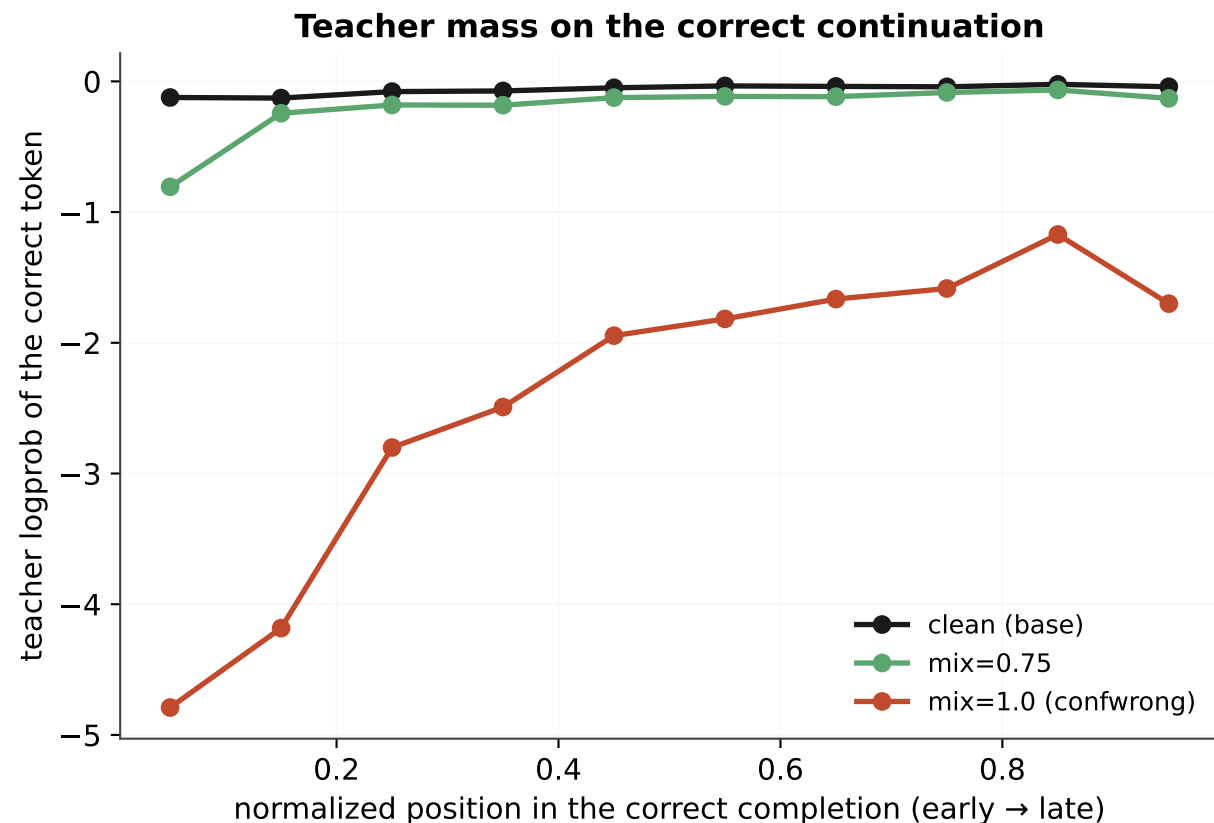
Refined conclusion: CONDITIONAL correct-mode survival

- OPD scores the student's on-policy CORRECT trajectories. Conditioned on a correct prefix, even the mix=0.75 teacher favors correct continuations — its constant-wrong mode is an unconditional / early-token phenomenon. Sharpening amplifies the argmax GIVEN that correct context, which is still correct.
- So neither low accuracy (dose-response) nor low entropy (this sweep) transfers damage. OPD only flips to an attack when the teacher is wrong even CONDITIONAL on correct context (mix=1.0, trained wrong everywhere) — then the correct mode is gone in-context.

On-policy distillation is robust: the binding variable is conditional correct-mode survival, not teacher accuracy OR entropy.

Mechanism, measured: conditional correct-mode survival

Score the base model's own CORRECT trajectories under each teacher (no training). Teachers at natural temperature.



Conditioned on a correct prefix, mean probability the teacher places on the correct next token:

clean 0.94 · mix=0.75 0.81 · mix=1.0 0.09.

The mix=0.75 teacher dips ONLY at the first token (its unconditional constant-wrong mode), then tracks the clean teacher — the correct mode survives in-context, so OPD recovers (student held 0.83).

The mix=1.0 teacher stays low everywhere: the correct mode is extinguished even given a correct prefix, so OPD has nothing correct to mode-seek and the student collapses (0.02). This is the direct, training-free measurement behind the dose-response cliff and the temperature null result.

Takeaways

What the data shows

- H1 — OPD repairs the reasoning policy. Full recovery to the teacher ceiling on OOD, across all three damage levels, with no labeled data and no reward model.
- H2 — on-policy beats off-policy. OPD > clean-SFT on OOD in every condition (+7 to +23pp). Continuing clean SFT even degraded OOD for underwrite_050 (0.63->0.60) and _010 (0.71->0.67).
- This is the compounding-error / distribution-shift story made concrete: more off-policy SFT keeps you off-policy; sampling from the student and correcting on-policy repairs the policy.
- Capability, not memorization: wrong-answer adoption ~ 0 for every repaired model.
- Severity-independent: subtle (~ 10 pp) and catastrophic (~ 30 pp) damage both recover fully.
- Teacher is the whole ballgame (adversarial-teacher pages): flip the teacher to confidently-wrong and the SAME loop attacks a healthy student (0.81->0.02 OOD). The binding variable is teacher token-level ENTROPY, not accuracy.

Caveats

- 1 seed; Wilson CIs are $\sim \pm 9$ pp wide. Multi-seed needed to tighten claims.
- Single model / single task family (GSM-style math, Qwen2.5-Math-1.5B).
- Teacher == exact pre-damage model (best case). Weak (E3) and adversarial (E10) teachers now probed; standalone / cross-family teachers still untested.

Proposed follow-up experiments

Tighten the core claim

- E1. Multi-seed (3-5) sweep over all conditions — turns the bars into significance.
- E2. OPD dose-response: steps (10/25/50/100/200) and prompt-pool size (1/8/64/256).
TM claim that even ~ 1 prompt works — test the data-efficiency floor here.

Probe the mechanism / limits

- E3. Weak-teacher OPD: teacher = a clean-SFT model (not the pristine base).
Tests the 'student can only be as good as the teacher' ceiling.
- E4. Self-repair: teacher = an EMA / earlier checkpoint of the damaged student itself.
Does the model bootstrap recovery with no external pre-damage checkpoint?
- E5. Forward-KL vs reverse-KL repair: does mode-covering forward KL recover broader coverage (fewer rare-case failures) at the cost of sharpness?

Push toward a stronger result

- E6. Hybrid objective: OPD per-token KL + sparse correctness reward (research idea on combining distillation + environment reward). Does it exceed the teacher ceiling?
- E7. Cross-task transfer: repair on GSM-train prompts, eval on MultiArith / GSM-Hard — does OPD restore broad reasoning or just GSM8K-shaped problems?
- E8. Repair a DIFFERENT damage type (instruction-following / format poison) to show the loop is damage-agnostic, not GSM-specific.
- E9. Scale check: same protocol on Qwen2.5-Math-7B to confirm it isn't a small-model artifact.

Highest value next: E1 (seeds) for rigor, then E3 (weak teacher) for the most novel insight.